

AWS CLI, SDK, IAM Roles & Policies

EC2 Instance Metadata (IMDS)

- AWS EC2 Instance Metadata (IMDS) is powerful but one of the least known features to developers
- It allows AWS EC2 instances to "learn about themselves" **without using an IAM Role for that purpose.**
- The URL is <http://169.254.169.254/latest/meta-data>
- You can retrieve the IAM Role name from the metadata, but you CANNOT retrieve the IAM Policy.
- Metadata = Info about the EC2 instance
- Userdata = launch script of the EC2 instance
- Let's practice and see what we can do with it!

IMDSv2 vs. IMDSv1

- IMDSv1 is accessing <http://169.254.169.254/latest/meta-data> directly
- IMDSv2 is more secure and is done in two steps:
 1. Get Session Token (limited validity) – using headers & PUT

```
$ TOKEN=`curl -X PUT "http://169.254.169.254/latest/api/token"
-H "X-aws-ec2-metadata-token-ttl-seconds: 21600"`
```

2. Use Session Token in IMDSv2 calls – using headers

```
$ curl http://169.254.169.254/latest/meta-data/profile
-H "X-aws-ec2-metadata-token: $TOKEN"
```

MFA with CLI

- To use MFA with the CLI, you must create a temporary session
- To do so, you must run the **STS GetSessionToken** API call
- **aws sts get-session-token** --serial-number arn-of-the-mfa-device --token-code code-from-token --duration-seconds 3600

```
{
  "Credentials": {
    "SecretAccessKey": "secret-access-key",
    "SessionToken": "temporary-session-token",
    "Expiration": "expiration-date-time",
    "AccessKeyId": "access-key-id"
  }
}
```

AWS SDK Overview

- What if you want to perform actions on AWS directly from your applications code ? (without using the CLI).
- You can use an SDK (software development kit) !
- Official SDKs are...
 - Java
 - .NET
 - Node.js
 - PHP
 - Python (named boto3 / botocore)
 - Go
 - Ruby
 - C++

AWS SDK Overview

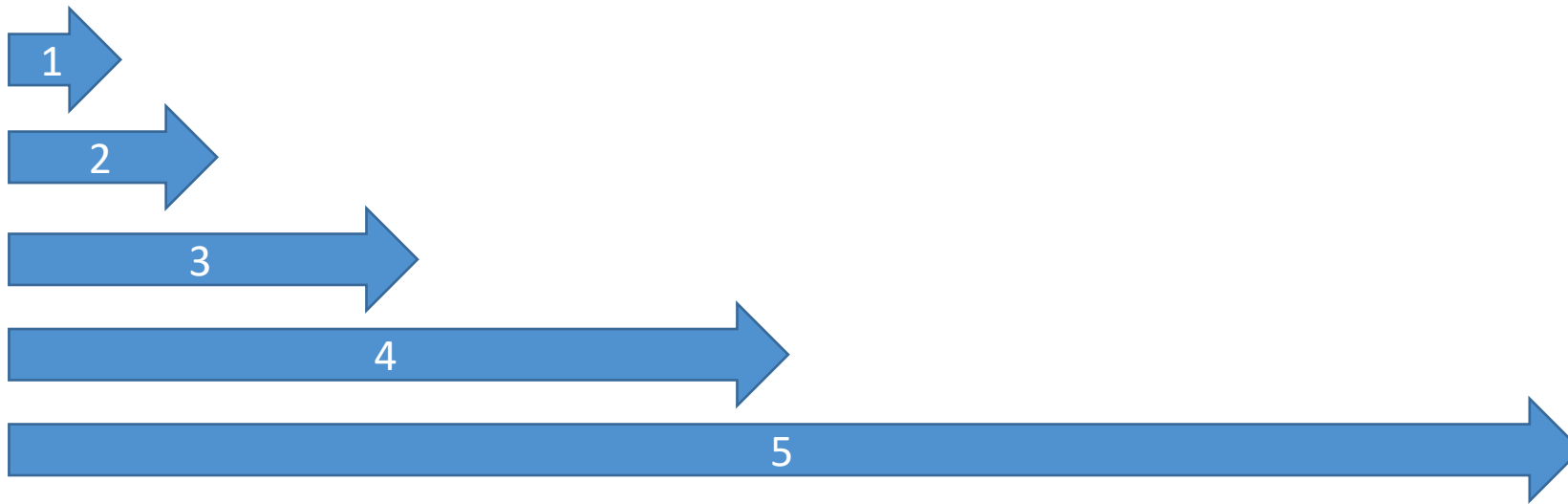
- We have to use the AWS SDK when coding against AWS Services such as DynamoDB
- Fun fact... the AWS CLI uses the Python SDK (boto3)
- The exam expects you to know when you should use an SDK
- We'll practice the AWS SDK when we get to the Lambda functions

AWS Limits (Quotas)

- API Rate Limits
 - **DescribeInstances** API for EC2 has a limit of 100 calls per seconds
 - **GetObject** on S3 has a limit of 5500 GET per second per prefix
 - For Intermittent Errors: implement Exponential Backoff
 - For Consistent Errors: request an API throttling limit increase
- Service Quotas (Service Limits)
 - Running On-Demand Standard Instances: 1152 vCPU
 - You can request a service limit increase by **opening a ticket**
 - You can request a service quota increase by using the **Service Quotas API**

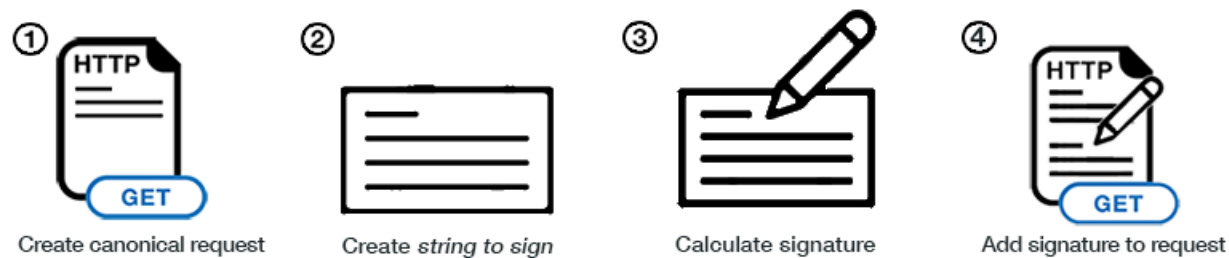
Exponential Backoff (any AWS service)

- If you get **ThrottlingException** intermittently, use exponential backoff
- Retry mechanism already included in AWS SDK API calls
- Must implement yourself if using the AWS API as-is or in specific cases



Signing AWS API requests

- When you call the AWS HTTP API, you sign the request so that AWS can identify you, using your AWS credentials (access key & secret key)
- Note: some requests to Amazon S3 don't need to be signed
- If you use the SDK or CLI, the HTTP requests are signed for you
- You should sign an AWS HTTP request using Signature v4 (SigV4)



SigV4 Request examples

- HTTP Header option (signature in Authorization header)

```
GET https://iam.amazonaws.com/?Action=ListUsers&Version=2010-05-08 HTTP/1.1
Authorization: AWS4-HMAC-SHA256 Credential=AKIDEXAMPLE/20150830/us-east-1/iam/aws4_request,
SignedHeaders=content-type;host;x-amz-date,
Signature=5d672d79c15b13162d9279b0855cfba6789a8edb4c82c400e06b5924a6f2b5d7
content-type: application/x-www-form-urlencoded; charset=utf-8
host: iam.amazonaws.com
x-amz-date: 20150830T123600Z
```

- Query String option, ex: S3 pre-signed URLs (signature in X-Amz-Signature)

```
GET https://iam.amazonaws.com?Action=ListUsers&Version=2010-05-08&
X-Amz-Algorithm=AWS4-HMAC-SHA256&
X-Amz-Credential=AKIDEXAMPLE%2F20150830%2Fus-east-1%2Fiam%2Faws4_request&
X-Amz-Date=20150830T123600Z&X-Amz-Expires=60&X-Amz-SignedHeaders=content-type%3Bhost&
X-Amz-Signature=37ac2f4fde00b0ac9bd9eadeb459b1bbec224158d66e7ae5fcadb70b2d181d02 HTTP/1.1
content-type: application/x-www-form-urlencoded; charset=utf-8
host: iam.amazonaws.com
```